GP2PL: From Generalized Planning Evidence to Certified BDI Plan Libraries for Temporally Extended Goals

Anonymous submission

Abstract

Generalized planning learns reusable behavior for problem families, whereas Belief–Desire–Intention (BDI) agents execute plan libraries. Direct adaptation leaves a representation gap: singleton-goal evidence may omit modules for subsidiary achievements and does not certify temporal composition. We introduce GP2PL (Generalized Planning to Plan Libraries), which compiles such evidence and Planning Domain Definition Language (PDDL) action models into certified BDI plan libraries. GP2PL validates and lifts evidence, generates schema-derived modules for producible goals, selects a compact closed atomic core, and appends query controllers guided by deterministic finite automata for finite-trace temporal goals without relearning that core. For the generated candidate language and supported fragment, accepted branches and controllers satisfy conditional soundness guarantees; uncertified obligations are rejected. Across 16 domains, five independently seeded cores achieve 98.68% mean held-out coverage (1.29 percentage-point sample standard deviation). Recursive candidates improve paired atomic success by 10.42 percentage points, and preservation certification improves paired temporal success by 9.36 percentage points. All 475 translated specifications match their gold finite-trace languages, and the fixed-core temporal study produces independently validated accepting traces for all 1,228 bound queries.

1 Introduction

Classical PDDL planning computes an action sequence for one grounded problem (McDermott et al. 1998); generalized planning learns reusable behavior for a family of related problems (Srivastava et al. 2011; Chen et al. 2026). A Belief–Desire–Intention (BDI) interpreter uses a different representation: a plan library that connects events and belief contexts to actions and subsidiary goals (Rao 1996; Meneguzzi and De Silva 2015). Turning a generalized policy into such a library is therefore a compilation problem, not a change of syntax.

In Blocks World, a learned rule for the lifted achievement `on(X,Y)` may use the action `stack(X,Y)` when the robot is holding X and the destination block is clear. If Y is not clear, however, an executable BDI library also needs a reusable plan for making it clear, even when that condition never appeared as a training goal. More generally, variables must remain safely bound, subsidiary goals must resolve to library plans, and recursive plans must make progress. Figure 1 summarizes the representation bridge and its query-reuse boundary.

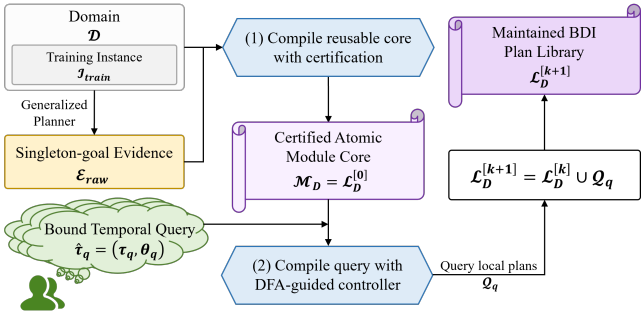

Figure 1: GP2PL separates reusable domain compilation from query-specific temporal compilation. An external generalized-planning backend derives raw singleton-goal evidence $\mathcal{E}_{\text{raw}}$ from domain D and training instances $\mathcal{I}_{\text{train}}$. Validated policy lifting compiles these inputs once into $\mathcal{M}_D = \mathcal{L}_D^{[0]}$; DFA-guided temporal compilation maps each bound query $\widehat{\tau}_q$ to query-local plans $\mathcal{Q}_q$, updating the sole maintained library by $\mathcal{L}_D^{[k+1]} = \mathcal{L}_D^{[k]} \cup \mathcal{Q}_q$.

Temporally extended goals add a second representation gap (De Giacomo and Vardi 2013; De Giacomo and Rubin 2018): a request to place one block on another and only later clear a third constrains the whole finite trace, and a conjunctive milestone must not be undone by a later plan—a form of BDI goal interference (Thangarajah, Padgham, and Winikoff 2003). Safe reuse therefore requires temporal control over the fixed domain library and observation after primitive actions.

The central design choice is reuse: temporal requests extend one maintained library instead of triggering a new domain-level learning run. Both compiler stages reason over typed action schemas and effects rather than domain names. Figure 2 exposes their internal transformations without mixing them with a domain-specific worked example.

GP2PL makes three contributions: it compiles singleton-goal evidence and PDDL schemas into an internally closed

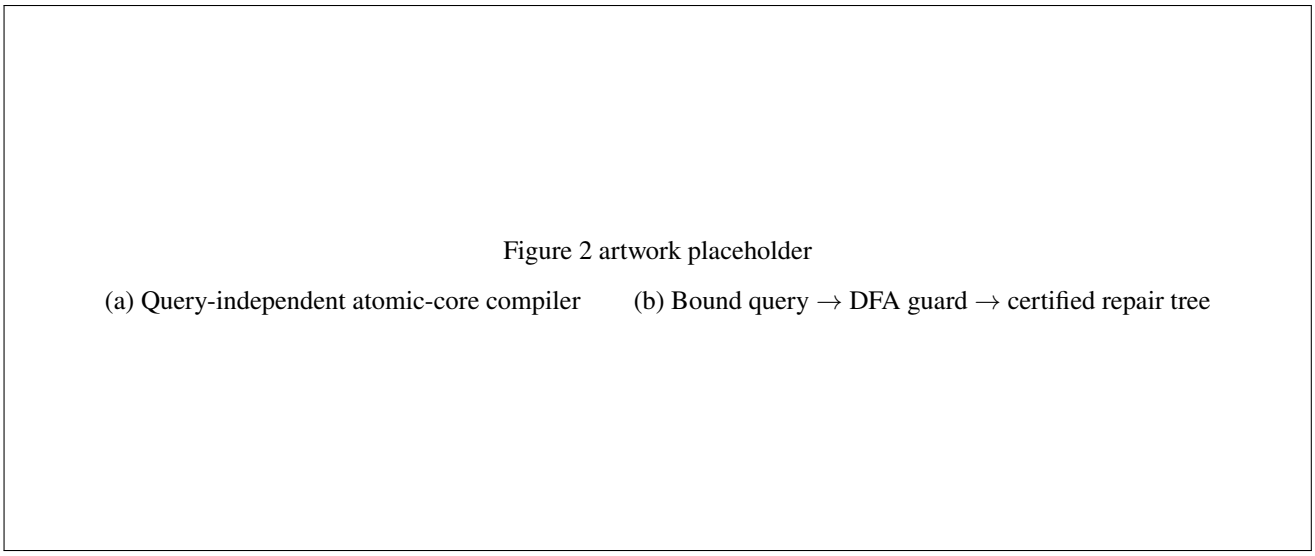

Figure 2: Inside the two GP2PL compiler stages. The query-independent stage normalizes and lifts evidence, constructs and certifies schema-derived candidates, and selects $\mathcal{M}_D = \mathcal{L}_D^{[0]}$. For the bound Blocks World tower query, the query-specific stage reuses selected completion summaries to certify a preservation portfolio $\mathbf{\Pi}_\chi$ and serialization ℓ_χ for its MONA-derived conjunctive progress guard, then renders the certified order as a balanced transition-repair tree. The resulting $\mathcal{Q}_q$ is appended without relearning the core. Figure 3 separately illustrates execution of the selected atomic core.

BDI core with modules for omitted producible subgoals; it accepts only branches carrying certificates for binding, symbolic execution, achievement, completion effects, and progress; and it appends preservation-safe finite-trace controllers over the fixed core, monitoring DFA progress after each primitive action.

The evaluation separates component effects from complete-system behavior through five-seed atomic evaluation, paired compiler ablations, and fixed-core temporal validation. These claims concern the declared candidate and formula fragments, not complete planning for arbitrary PDDL–LTL_f products. Formal definitions, proofs, and the reporting protocol appear in the Technical Supplement.

2 Problem Formulation and Foundations

2.1 Planning Domains and Generalized Evidence

We use typed STRIPS PDDL plus a bounded-integer resource fragment (Fikes and Nilsson 1971; McDermott et al. 1998; Fox and Long 2003). A domain is $D = \langle \mathcal{P}, \mathcal{F}, \mathcal{A}, \mathcal{T} \rangle$ with predicate schemas, integer-valued resource functions, action schemas, and a type hierarchy; a state $s = (s_\mathcal{P}, s_\mathcal{F})$ combines Boolean and integer valuations under the standard add/delete successor. The numeric component admits finite integer values, comparisons with constants, and constant integer changes; arbitrary arithmetic and continuous change are excluded. A problem instance is $I = \langle D, O_I, s_I^0, G_I \rangle$ with typed objects, initial state, and PDDL achievement condition, and a generalized-planning task supplies finite training and test instance sets sharing D. MOOSE applies goal regression (Fikes, Hart, and Nilsson 1972; Alcázar et al. 2013) to singleton goals and lifts the plans into first-order condition-to-macro rules (Chen et al. 2026). We treat these rules as evidence: they demonstrate ways to achieve observed singleton goals, but need not constitute a closed or compact atomic module core.

2.2 BDI Plan Libraries in AgentSpeak(L)

A procedural BDI plan rule has a triggering event, an applicability context, and a body (Meneguzzi and De Silva 2015). In the AgentSpeak(L) realization evaluated here (Rao 1996), an achievement plan is

$$+!p(\bar{X}) : C \leftarrow b_1; \ldots; b_m,$$

where C is a conjunction of beliefs and comparisons and each b_i is a PDDL action or an internal achievement call $!q(\bar{Y})$. A library is lifted when plans contain variables rather than training objects, and internally closed when every internal call resolves to a type-compatible selected module; closure is a signature-resolution property, not per-state applicability.

Our certificates apply to this trigger–context–body model and to the PDDL semantics of primitive actions. Formally, a candidate branch is $b = \langle g_b, C_b, \beta_b, \Sigma_b, \pi_b \rangle$: one lifted achievement literal, a conjunctive applicability condition, a finite body of primitive actions and internal calls, a conservative conditional module-completion summary, and provenance. We map selected branches to AgentSpeak(L) and evaluate them with Jason (Bordini, Hübner, and Wooldridge 2007); claims about another BDI language would require a semantics-preserving translation and an independent execution study. Query controllers follow declarative goal patterns by rechecking their intended state after subsidiary plans return (Hübner, Bordini, and Wooldridge 2006).

2.3 Finite-Trace Temporal Goals

Linear temporal logic over finite traces (LTL_f) is interpreted over a nonempty finite state sequence (De Giacomo and Vardi

2013). For proposition alphabet AP_q, the declared input language Φ_{syn} admits literals, conjunction, F, X, and strong U over AP_q, with negation restricted to atoms; the full grammar and finite-trace semantics appear in the Technical Supplement, Sec. 1. The evaluation family $\Phi_{\text{bench}} \subset \Phi_{\text{syn}}$ contains instances of the five predeclared profiles in Sec. 7; membership in either set is not an action-strategy guarantee.

Following automata-theoretic planning (De Giacomo and Rubin 2018), we construct the deterministic finite automaton (DFA) $\mathcal{D}_q = \langle Q_q^{\text{dfa}}, 2^{AP_q}, \delta_q, q_q^0, F_q \rangle$ with LTLf2DFA and MONA (Fuggitti 2020; Henriksen et al. 1995), and a valuation map val_q interprets a PDDL state over AP_q. For automaton state z, let $d_q(z)$ be the shortest directed-path length to F_q, or ∞ when no accepting state is reachable. A progress transition (q_s, χ, q_t) satisfies $d_q(q_t) < d_q(q_s) < \infty$, and a supported guard χ has normal form

$$\chi = \bigwedge_{G \in P_\chi^+} G \wedge \bigwedge_{N \in P_\chi^-} \neg N, \qquad \mathcal{O}_\chi = \langle P_\chi^+, P_\chi^- \rangle,$$

where the signed transition obligation $\mathcal{O}_\chi$ separates positive achievements from atoms that must remain absent. The certificate-dependent subset $\Phi_{\text{cert}}(D, \mathcal{M}_D) \subseteq \Phi_{\text{syn}}$ contains the validated bound formulas whose required DFA progress obligations admit the certificates of Sec. 5; soundness claims apply to this subset, empirical claims to accepted members of Φ_{bench}. A query-local monitor evaluates the DFA initially and after every successful primitive PDDL action.

2.4 Certified Compilation Problem

Compilation problem. Given D, training instances $\mathcal{I}_{\text{train}}$, and normalized singleton-goal evidence E, atomic compilation produces the certified atomic module core $\mathcal{M}_D$: the query-independent branches whose triggers are singleton achievements.

An unbound temporal specification is τ_q, its externally supplied type-consistent object binding is θ_q, and $\widehat{\tau}_q = (\tau_q, \theta_q)$ is the validated bound query. Temporal compilation produces the query-local plan set $\mathcal{Q}_q$, comprising its top-level goal, DFA dispatch, and transition-repair plans. For appended queries $q_1, \ldots, q_k$, the one maintained domain library is

$$\mathcal{L}_D^{[0]} = \mathcal{M}_D, \qquad \mathcal{L}_D^{[k]} = \mathcal{M}_D \cup \bigcup_{i=1}^{k} \mathcal{Q}_{q_i}.$$

Thus $\mathcal{M}_D$ is the stable logical core of $\mathcal{L}_D^{[k]}$, not a second persisted library, and appending $\mathcal{Q}_{q_i}$ neither reconstructs nor reoptimizes the atomic core. Throughout, calligraphic symbols denote finite sets or structured artifacts; b denotes one candidate branch, e one normalized evidence rule, and χ one DFA transition guard. The Technical Supplement, Sec. 1 gives the formal definitions, supported-fragment assumptions, and the acceptance and rejection boundary used below.

3 Related Work and Positioning

Generalized planning. Generalized planning seeks compact policies or programs that solve families of planning instances (Jiménez, Segovia-Aguas, and Jonsson 2019; Srivastava et al. 2011; Bonet and Geffner 2018; Francès, Bonet, and Geffner 2021; Bonet and Geffner 2024), via policy sketches and answer-set optimization (Drexler, Seipp, and Geffner 2022, 2024), lifted decision lists (Yang et al. 2022), or terminating programs (Segovia-Aguas, E-Martín, and Jiménez 2022); MOOSE learns lifted condition-to-macro rules by singleton-goal regression (Chen et al. 2026). Recent BDI integrations invoke a classical planner at run time (Xu et al. 2024) or use generalized planning for means–ends reasoning (Meneguzzi, Fraga Pereira, and Oren 2025). GP2PL instead addresses the representation-compilation problem: transforming singleton-goal evidence and action-schema-derived candidates into a query-independent atomic module core $\mathcal{M}_D$ that later queries extend.

Modular policies and BDI plan libraries. Policy-reuse languages let one generalized policy invoke another with parameters (Bonet, Drexler, and Geffner 2024), motivating internal calls such as `!clear(Y)`. Procedural BDI languages organize behavior as event-triggered plans selected from a context-dependent library (Meneguzzi and De Silva 2015; De Silva, Meneguzzi, and Logan 2020; Rao 1996; Bordini, Hübner, and Wooldridge 2007), with declarative-goal rechecking (Hübner, Bordini, and Wooldridge 2006), plan-failure handling (Sardina and Padgham 2007), and effect-summary interference analysis (Thangarajah, Padgham, and Winikoff 2003). GP2PL adopts these execution principles and derives certified modules, completion summaries, and declarative completion controllers from planning evidence and PDDL schemas.

Finite-trace temporal control. LTL_f admits automata-based reasoning (De Giacomo and Vardi 2013), which we instantiate with the MONA-backed LTLf2DFA construction (Fuggitti 2020; Henriksen et al. 1995). Full temporal synthesis solves the domain–automaton product (De Giacomo and Rubin 2018); GP2PL instead composes certified repairs over an existing BDI library and makes no complete product-synthesis claim. Temporally extended goals also have explicit agent-language semantics (Hindriks, van der Hoek, and van Riemsdijk 2009); we retain standard plan execution and add a query-local monitor. Following the fixed-planner principle of (Bonassi et al. 2023), FOND4LTLf (De Giacomo and Fuggitti 2021) is a task-level reference on its accepted subset, not a compiler ablation.

Natural-language temporal specifications. Prior systems map natural-language instructions to temporal logic through pattern libraries (Fuggitti and Chakraborti 2023), lifted language-to-logic translation (Chen et al. 2023; Guo, Kingston, and Kavraki 2025), or modular grounding (Liu et al. 2023); prompt-based parsers remain sensitive to input perturbations (Beurer-Kellner, Fischer, and Vechev 2023; Zhuo et al. 2023). Translation is an evaluated input condition here, not a contribution: we validate typed, externally bound parametric LTL_f by exact finite-trace language equivalence before GP2PL compilation, with the protocol and frozen prompt in the Technical Supplement, Sec. 4.

Position of this work. Prior work supplies generalized policies, answer-set optimization, AgentSpeak execution semantics, and LTL_f automata as separate artifacts. GP2PL formulates their connection as a certificate-carrying compilation problem that establishes schema-level obligations before emitting one maintained executable library, echoing proof-carrying code (Necula 1997) with narrower symbolic planning obligations. Goal regression, answer-set solving, AgentSpeak semantics, and LTL_f-to-DFA translation remain prior components.

4 Certified Atomic-Module Compilation

4.1 Method Overview

GP2PL has two certification boundaries. Domain compilation maps $D, \mathcal{I}_{\mathrm{train}}, \mathcal{E}_{\mathrm{raw}}$ to $\mathcal{M}_D$ by verifying $\mathrm{Cert}_D(b)$ and recording Σ_b before feasible-core selection. Query compilation reuses the selected summaries to certify $(\boldsymbol{\ell}_\chi, \boldsymbol{\Pi}_\chi)$ for each DFA guard χ, without relearning $\mathcal{M}_D$. The Technical Supplement, Sec. 3 gives the full pseudocode.

All obligations are checked on trigger–context–body rules before rendering. The AgentSpeak(L) rendering preserves each selected trigger, context, ordered body, and internal-call edge. The Technical Supplement, Sec. 1 gives the formal accept/reject boundary.

4.2 Evidence Normalization and Canonical Lifting

An evidence provider is admissible when its raw artifact $\mathcal{E}_{\mathrm{raw}}$ parses to a singleton-goal evidence program E_0. Canonical lifting produces $E = \mathrm{Lift}_D(E_0)$, whose rules pair a partial state condition and one goal atom with a PDDL action macro, numeric effects, and provenance. Every symbol and arity must occur in D. Lifting capture-avoidingly alpha-normalizes provider variables, preserves repeated-variable sharing, and keeps constants declared by D fixed, so `on(block0,block1)` becomes `on(X,Y)`. It normalizes a policy already generalized by the provider rather than inferring one from ground plans. Atomic evidence is restricted to positive singleton achievements; negative literals remain guard conditions and do not induce synthetic atomic goals.

4.3 Action-Schema Candidate Generation

Evidence need not cover the signatures required by internal calls. Candidate generation therefore expands from evidence goals to schema-producible targets, independently of which goals the provider observed. Let $\mathrm{goalPred}(E)$ contain the predicate symbols targeted by positive predicate evidence and $\mathrm{prod}(D)$ the predicates occurring in some action's add effects; the domain-wide producible target universe is

$$T_D(E) = \mathrm{goalPred}(E) \cup \mathrm{prod}(D).$$

Every add-effect predicate thus receives candidate atomic modules. Predicates with neither validated evidence nor a positive producer do not; static and delete-only predicates remain non-achievement context. Supported numeric equalities form $T_D^{\mathbb{Z}}(E)$, the admitted singleton integer-equality targets in E.

The finite grammar $\mathfrak{G}$ contains six constructors—producer, regression, recursion, precondition repair, resource discharge, and numeric progress. For Boolean and numeric targets, LiftSchemas computes $\mathcal{C}_D(E) = \mathrm{Inst}_D(\mathfrak{G}, T_D(E), T_D^{\mathbb{Z}}(E))$, using only typed schemas and effects from D, then alpha-normalizes each variable-level branch. For target set T, the producer–precondition graph $\mathcal{G}_D^{\mathrm{prod}}(T)$ has an edge $p \to r$ when a producer for p has a positive dynamic precondition with predicate r. Backward STRIPS regression (Fikes, Hart, and Nilsson 1972; Alcázar et al. 2013; Chen et al. 2026) on an acyclic reachable graph first unifies compatible open requirements, introduces fresh variables only for unbound parameters, and rejects conflicting deletes. An alpha-normalized regression state records open requirements, producer role, and resource modes; search stops before repeating such a state and keeps the shortest body for an equivalent completion summary. Construction is therefore finite without an arbitrary action-depth constant; validated evidence macros are never truncated. Symbolic progression verifies every candidate, while cyclic dependencies require provider evidence or a separately certified recursive module. Let $\mathcal{C}_E$ be the validated evidence branches. The resulting candidate set $\mathcal{C}_{D,E} = \mathcal{C}_E \cup \mathcal{C}_D(E)$ is finite, but $\mathfrak{G}$ is not a complete hypothesis language for PDDL planning. Types constrain unification; rendered sort membership is context-only.

4.4 Candidate Soundness Certificates

Each candidate's conditional module-completion summary Σ_b has Boolean projections $\mathrm{MustAdd}_b$, MayAdd_b, and $\mathrm{MayDelete}_b$, together with any exact constant-integer delta and keyed resource-mode transition. The $\mathrm{MustAdd}_b$ facts hold after every feasible completion; MayAdd_b and $\mathrm{MayDelete}_b$ overapproximate possible effects. Selected branches also satisfy typed binding, symbolic executability, and target achievement; $\mathrm{Cert}_D(b)$ denotes their conjunction, and the core satisfies $\mathrm{Closed}_D(\mathcal{M}_D)$. The Technical Supplement, Sec. 1 tabulates constructor-specific obligations; the load-bearing certificates follow.

Binding, symbolic execution, and internal-call closure. Action and subgoal variables must be bound by the trigger, positive context, or certified static sort membership; negative contexts and comparisons only filter. Symbolic progression proves preconditions and target achievement while recording $\mathrm{MustAdd}$, MayAdd, and $\mathrm{MayDelete}$. $\mathrm{Closed}_D(\mathcal{R})$ holds when every internal call in plan set $\mathcal{R}$ type-unifies with a selected trigger under D's type hierarchy. Requiring $\mathrm{Closed}_D(\mathcal{M}_D)$ is a set-level Clingo obligation: it proves signature resolution, not branch applicability in every reachable state.

Well-founded recursive progress. A same-predicate recursive branch b requires a ranking function $\rho_b : \mathrm{States}(D) \to \mathbb{N}$, a non-negative relation or anchored acyclic-cone count that strictly decreases, together with an already-satisfied, direct-producer, or validated-evidence terminal branch. Threats are checked over the preparation graph selected by Clingo; a branch may add the ranked relation outside the protected cone only under a guard proving the

new anchor differs from the protected one. The ranking is a compile-time witness, not an agent belief, and claims only generated schema-certified features. The construction follows relational progress witnesses in general-policy work (Francès, Bonet, and Geffner 2021).

Target-preserving resource discharge. Effects, not predicate names, identify temporary occupation of a reusable resource: when an acquisition action deletes an availability fact $free(R)$ and adds an occupancy relation $held(R, U)$ (names illustrative), the compiler infers the resource and occupant arguments from the effect structure and forms a finite graph of alpha-normalized keyed occupancy modes. A discharge body is accepted only if symbolic execution removes the current occupancy mode, restores the availability fact, leaves the atomic target true, and does not recreate an earlier normalized mode; excluding repeated modes makes the search finite without assuming that release is one action.

Ranked cross-module precondition repair. Preparing one of several dynamic preconditions may temporarily change another, such as an object's location. A repair branch may defer an unrelated sibling only through a schema-inferred single-valued fluent when every deferred sibling has one producer, all static feasibility witnesses remain, and the original target is retried before primitive production. For a candidate core $\mathcal{M}$, Clingo assigns each trigger signature in its selected call graph a dependency rank $\kappa_{\mathcal{M}}$ that strictly decreases along every selected call edge; same-predicate recursion instead requires the relational certificate above.

4.5 Feasible-Core Optimization

Validated evidence macros and certified action-schema-derived candidates enter one answer-set optimization problem solved with Clingo (Gebser et al. 2019). Evidence rules induce coverage obligations $\mathrm{covers}_D(b, e)$. Coverage requires alpha-equivalent triggers and either an equivalent body and summary, the same body under a weaker context, or a producer refinement with the same $\mathrm{MustAdd}$, no additional $\mathrm{MayDelete}$, and equivalent numeric and keyed-resource summaries. For each nonrecursive schema branch, $\mathrm{realizes}_D(b, c)$ permits an equivalent branch, a refinement with no weaker completion contract, or a target-preserving resource-discharge alternative with the same achievement contract. Body-prefix similarity alone satisfies neither relation. Define

$$\mathcal{C}^{\checkmark}_{D,E} = \{b \in \mathcal{C}_{D,E} \mid \mathrm{Cert}_D(b)\},$$
$$\mathfrak{F}_{D,E} = \{\mathcal{M} \subseteq \mathcal{C}^{\checkmark}_{D,E} \mid \mathcal{M} \models \Omega_{D,E} \wedge \mathrm{Closed}_D(\mathcal{M})\},$$

where $\Omega_{D,E}$ contains evidence coverage, schema-achievement coverage, preparation acyclicity, and recursive-ranking compatibility, and resource discharge is part of the per-branch predicate Cert_D. The optimizer uses the lexicographic objective

$$\mathcal{M}_D = \operatorname*{lexargmin}_{\mathcal{M} \in \mathfrak{F}_{D,E}} \big\langle -N_{\mathrm{rel}}(\mathcal{M}), -N_{\mathrm{prep}}(\mathcal{M}), |\mathcal{M}|, N_C(\mathcal{M}), N_\beta(\mathcal{M}) \big\rangle,$$

where N_{rel} and N_{prep} count compatible relation-ranked recursion and acyclic precondition-repair capabilities, N_C counts context literals, and N_β counts body steps. The selected trigger–context–body rules, rather than their AgentSpeak rendering, define $\mathcal{M}_D$; the claim is global only within $\mathcal{C}^{\checkmark}_{D,E}$, not over ungenerated programs.

Figure 3 isolates how the selected atomic core $\mathcal{M}_D$ is instantiated in a Blocks World state not encoded in its rules.

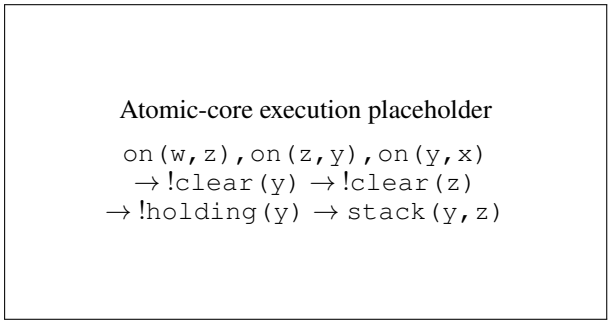

Figure 3: Execution of the selected atomic core in Blocks World. From `on(w,z),on(z,y),on(y,x)`, the goal `!on(y,z)` calls `!clear(y)`. The same `clear/1` module recurs on `z`, decreasing the certified obstruction count before `holding/1` and the direct `stack(y,z)` producer complete the goal. The selected rules contain variables rather than these illustrative object names.

5 DFA-Guided Composition of Temporally Extended Goals

Progress-edge compilation. Given the validated bound query $\widehat{\tau}_q$ and DFA $\mathcal{D}_q$ of Sec. 2, GP2PL compiles one controller for each progress edge (q_s, χ, q_t); an accepting `true` self-loop needs none. The external binding θ_q instantiates query propositions while the shared atomic core $\mathcal{M}_D$ remains lifted. For signed obligation $\mathcal{O}_\chi = \langle P^+_\chi, P^-_\chi \rangle$, compilation derives a certified serialization $\boldsymbol{\ell}_\chi$ and occurrence-specific preservation portfolios $\mathbf{\Pi}_\chi$ from the selected completion summaries. Failure to certify this pair rejects the progress edge; otherwise $\boldsymbol{\ell}_\chi$ is rendered as the balanced binary transition-repair tree of Sec. 5.3.

Figure 2(b) instantiates this construction for a bound Blocks World tower query. Its DFA progress guard requires three `on` literals to hold together; conditional completion summaries induce a bottom-up preservation-safe order, and the balanced transition-repair tree emits $\mathcal{Q}_q$ for append to $\mathcal{L}^{[k]}_D$.

5.1 Summaries and Preservation Portfolios

A relational fixed point over selected calls computes each branch summary Σ_b, following the definite/potential effect distinction in BDI goal-interference analysis (Thangarajah, Padgham, and Winikoff 2003). For positive occurrence $G_i \in P^+_\chi$, a preservation portfolio $\Pi_{\chi,i} \subseteq \mathcal{M}_D$ contains the certified branches allowed to establish G_i under the obligations protected at that occurrence. Portfolios are occurrence-specific: two occurrences of the same predicate may admit different branches.

A negative guard $\neg N$ is observed absent or discharged by an exact certified deleter; it never becomes a synthetic $!\neg N$ goal. A portfolio branch is admissible only when its conditional MayAdd excludes N. For a chosen portfolio tuple, the threat-induced precedence graph $\mathcal{G}_\chi = (P_\chi^+, \prec_\chi)$ places G_j before G_i when a feasible $\mathrm{MayDelete}$ of $\Pi_{\chi,j}$ may delete G_i. An acyclic graph yields a topological serialization. If no portfolio choice removes a cycle, compilation requires a joint branch b_χ^{joint} whose complete net effect establishes the entire guard, is callable from every positive occurrence it establishes, and retains every query variable; otherwise it rejects the edge. The exact fixed point, precedence relation, and cycle conditions appear in the Technical Supplement, Secs. 1–2.

5.2 Numeric and Joint Guard Certificates

Numeric obligations reuse the same certificate machinery. A positive integer equality is achieved by a complete constant-delta action: a unit delta requires strict direction and a larger delta the exact predecessor value. A mixed Boolean–numeric guard uses complete action-only net effects; for example, a body for $at(P, L) \wedge fuel(V) = 3$ must symbolically establish $at(P, L)$ and the exact integer value 3 in its final state while preserving every other signed obligation. Negated equality is observation-only without a certified change-away action. Numeric preparation uses $\rho_{\mathrm{num}} = \langle n_{\mathrm{miss}}, d_{\mathrm{target}} \rangle$, ordering missing preconditions before bounded target distance. Strong-Until source invariants are the signed literals common to every non-progress self-loop guard and are preserved until progress (Technical Supplement, Sec. 2).

5.3 Balanced Binary Transition-Repair Trees

For query q, let $\mathcal{R}_{q_s,\chi}$ denote the plans for one source-state transition obligation; the query-local plan set $\mathcal{Q}_q$ contains the top-level dispatcher together with every accepted $\mathcal{R}_{q_s,\chi}$, one per progress obligation. Each $\mathcal{R}_{q_s,\chi}$ realizes $\boldsymbol{\ell}_\chi$ by recursively splitting contiguous literal ranges at their midpoint. Internal nodes dispatch two child ranges, leaves observe or repair one signed literal, and the completion plan retries only while the monitor remains at q_s. The certificate pair, rather than the tree shape, provides the temporal guarantee. Trigger fan-out denotes the number of sibling AgentSpeak plans sharing one transition-repair trigger. Flat control therefore has fan-out $O(n)$, whereas a direct linear body gives an $O(n)$-long intention continuation; balanced dispatch preserves the certified order while bounding both local quantities by two. The construction and suffix-safety argument appear in the Technical Supplement, Sec. 2.

Appending $\mathcal{Q}_q$ updates $\mathcal{L}_D^{[k]}$ as defined in Sec. 2 without changing $\mathcal{M}_D$. Query-local goals are control symbols, not PDDL fluents or exported actions; only primitive PDDL actions change the observed state.

6 Conditional Guarantees

The guarantees concern accepted artifacts under their stated premises. They are conditional on the generated certified candidate set and certificate-accepted temporal subset; they do not assert complete candidate generation, universal AgentSpeak optimality, or complete strategy synthesis for arbitrary PDDL domains and DFAs.

Atomic branch soundness. Proposition 1. If an emitted $p(\bar{X})$ branch's context holds under a type-consistent grounding and called modules satisfy their certified achievement and completion summaries, the branch is executable and returns with $p(\bar{X})$. Binding grounds each term; induction over symbolic progression establishes preconditions and final achievement; recursion uses its ranking and preparation rechecks the producer context. □ The proposition establishes local soundness, not reachable-state coverage.

Certified-candidate-set optimality. Proposition 2. For the grounded selection instance defined above, a Clingo optimum $\mathcal{M}_D$ belongs to $\mathfrak{F}_{D,E}$ and minimizes the declared lexicographic cost over that feasible family. The Technical Supplement proves a bidirectional correspondence between answer sets and feasible subsets; the result does not imply complete candidate generation or optimality over all AgentSpeak programs.

Supported transition-composition soundness. Proposition 3. Let $\mathcal{O}_\chi = \langle P_\chi^+, P_\chi^- \rangle$ be an accepted signed transition obligation. If each required signed repair has an applicable certified branch, each selected call terminates, and later repairs preserve earlier obligations, then at the end of a complete transition-repair pass the controller retries exactly when the monitor reports the source state; otherwise it returns to dispatch at the current state. Every intervening state change is an actual DFA edge enabled by the observed valuation, and a rejecting state cannot report success.

Proof sketch. Induct over the certified order: each leaf establishes its signed literal; later summaries exclude deleting prior positives or adding forbidden atoms. If the source is left during a call, the remaining suffix still contains only these certified calls and observations, so it preserves the prefix at return boundaries. The monitor evaluates full cubes after every action and cannot be written by the controller; the suffix can therefore cause only real, fully observed DFA transitions. After the full pass, the completion plan tests the current state and either retries or returns to top-level dispatch. □ This does not imply complete action-strategy synthesis or guarantee that a post-exit suffix avoids a rejecting state.

Structural and monitor-fidelity guarantees. Three further guarantees are stated and proved in the Technical Supplement, Sec. 2. The balanced tree for n signed literals and e repair plans has $2n + e + 2$ plans, fan-out and body length at most two, and depth $O(\log n)$ while preserving the certified order (Proposition 4). After every successful primitive action, the runtime monitor state equals the deterministic automaton state over the observed finite trace, so leaving a controller source state is exactly a DFA exit and controller plans cannot write monitor beliefs (Proposition 5). If the initial valuation is accepting, successful execution commits no primitive action and denotes the one-state trace $\langle s_0 \rangle$ rather than an inserted noop, which could change strong-Next or Until truth (Proposition 6).

Complete proofs, including the binding, symbolic-progression, relational-ranking, resource-discharge, and tree-order lemmas used by these propositions, appear in the Technical Supplement, Sec. 2.

7 Experimental Evaluation

7.1 Questions and Comparisons

Atomic compilation. The cumulative comparison begins with Evidence Only, which validates and retains provider macros. Direct Producers adds action-only producers for the producible target universe. Maximum Feasible adds certified recursive preparation and target-preserving resource discharge, then retains a maximum-cardinality feasible subset. Full GP2PL applies the lexicographic Clingo objective to that same certified candidate set. These rows isolate target expansion, recursive candidate generation, and compact selection.

Temporal composition. Unprotected Serialization uses the same automaton, library, and primitive-step monitor without completion-effect analysis. Module-Return Monitor retains completion-effect ordering, preservation portfolios, and balanced repair, but checks the DFA only after an invoked atomic achievement module finishes all primitive actions and returns control to the query controller. Certified Balanced instead checks the DFA after every primitive action. The two references expose the effects of unprotected serialization and a coarser observation boundary.

7.2 Benchmarks and Protocol

Corpus and repetitions. The corpus contains eight classical and four bounded-integer MOOSE domains with official splits (Chen et al. 2026; Chen 2026), plus four predeclared serialized-width families (Lipovetzky and Geffner 2012): Blocksworld Clear and On (Francès, Bonet, and Geffner 2021; Bonet 2026), Tower (Bacchus 2001; Potassco 2026), and Depots (Bonet and Geffner 2018; RLeap Project 2026). MOOSE readable policies provide $\mathcal{E}_{\text{raw}}$. Following its protocol, the atomic study uses every training instance and three goal orders to compile five independently seeded cores.

Temporal benchmark construction. For each held-out problem, benchmark construction ignores the original achievement goal. Predicate and bounded-integer changes instantiate five predeclared profiles: $F(A \wedge B)$, $F(A \wedge \neg B)$, $F(A \wedge X(F(B)))$, $F(A \wedge X(F(B \wedge X(F(C)))))$, and strong $A\,U\,B$. Selection retains one witness-backed query per held-out problem, yielding 1,228 queries over 475 distinct translation inputs.

Controlled inputs and validation. Atomic variants share each seed's evidence. Temporal variants share the same atomic core, binding, and DFA. Atomic success requires Jason completion and original-goal VAL (Bordini, Hübner, and Wooldridge 2007; Howey, Long, and Fox 2004). Translation success requires exact finite-trace language equivalence between predicted and gold DFAs. Temporal success requires controller compilation, Jason completion, neutral-goal VAL, and independent trace acceptance by both DFAs; all failures remain in the denominator. Endpoint comparisons are paired by held-out identifier, with seeded atomic outcomes aggregated within identifier before inference. Construction details, resource limits, cost breakdowns, and inference rules appear in the Technical Supplement, Secs. 4–5.

7.3 Paired Component Ablations

Method	Valid (%)	Branches	KiB
Evidence Only	88.3	835.0 ± 15.8	488.3 ± 8.3
Direct Producers	88.3	1153.0 ± 15.8	549.2 ± 8.3
Maximum Feasible	**98.7**	1533.0 ± 15.8	645.9 ± 8.3
Full GP2PL	**98.7**	**1499.0 ± 15.8**	**635.1 ± 8.3**

Table 1: Paired atomic compiler comparison over 6,140 held-out seed–case executions (16 domains, 1,228 cases, five fixed seeds). Branches and KiB (library size) are mean ± sample standard deviation across seeds. Bold marks tied best coverage; blue bold marks Full GP2PL and its smaller core at equal coverage.

All variants compile 80 libraries and cover all 365 producible targets except Evidence Only (90/365). Certified recursive candidate generation supplies the 10.42-percentage-point gain in Table 1, entirely on the four added serialized-width domains. After averaging the five seeded outcomes within each identifier, every nonzero difference favors the recursive variants. At equal coverage, the lexicographic objective selects the smallest certified library.

Method	Valid	PAR-2 s	Plans	Fan-out
Unprotected Serialization	1113/1228	342.0	7	3
Module-Return Monitor	1212/1228	53.1	13	2
Certified Balanced	**1228/1228**	5.6	13	**2**

Table 2: Paired temporal compiler comparison over 1,228 queries. PAR-2 charges failures twice the 1,800-second limit. Plans is median controller size and Fan-out is the maximum number of sibling plans sharing one repair trigger. Bold marks best coverage; blue bold marks selected Balanced, which bounds fan-out at two; selection is structural, not runtime-based.

Certified Balanced yields a 9.36-percentage-point gain over Unprotected Serialization in Table 2; the four bounded-integer domains account for 113 of the 115 gains, and Transport accounts for two. We select Balanced for its bounded local controller structure, not for runtime superiority; helper plans are its expected representational cost. Module-return observation reduces valid coverage, supporting the primitive-step observation boundary. This paired matrix conditions on its registered seed-0 core and does not estimate cross-seed temporal robustness.

7.4 Full-System Coverage and Robustness

Five-seed atomic robustness. Across five independently compiled atomic cores, mean held-out coverage is 98.68% ± 1.29 percentage points (Table 3). Variation is confined to Logistics and Depots, where cyclic producer and resource

dependencies exceed the current certificate class. Every reported success satisfies original-goal VAL, and no domain-specific rule was introduced after observing held-out outcomes.

Scope	Cases/seed	Valid/seed	Coverage (%)
All 16 domains	1,228	1,187–1,224	98.7 ± 1.3
All-seed complete (14)	1,127	1,127	100.0 ± 0.0
Logistics	90	54–90	86.4 ± 16.7
Depots	11	6–9	63.6 ± 11.1

Table 3: Full GP2PL atomic held-out coverage over $n = 5$ predeclared evidence seeds. Coverage is mean $\pm$ sample standard deviation; Valid/seed gives the minimum–maximum count. Domains complete in every seed are aggregated; others are shown separately.

End-to-end temporal validation. The five registered structures instantiate Φ_{bench} rather than exhausting Φ_{syn}, and execution includes only queries admitted to $\Phi_{\text{cert}}(D, \mathcal{M}_D)$. GPT-5.5 translates the registered specifications. All 475/475 controlled-language specifications are valid and DFA-language equivalent, and all 1,228 admitted bound queries complete and pass neutral-goal VAL and both DFA trace oracles. Runtime is 5.5 seconds median (interquartile range 4.28–8.49), and action count is 2 median (interquartile range 1–2).

This study fixes one preregistered seed-0 atomic core per domain; its result supports neither cross-seed claims nor arbitrary PDDL–LTL_f completeness.

7.5 Evidence-to-Library Comparison

Table 4 isolates the compiler's gain over its evidence provider on the four added families. Raw MOOSE executes each seed-specific policy directly; Full GP2PL compiles the same evidence with typed action schemas. Both use the same 740 held-out seed–case pairs.

Method	Valid	Coverage (%)
Raw MOOSE evidence	117/740	15.8
Full GP2PL	**720/740**	**97.3**

Table 4: Same-scope evidence-to-library comparison on 740 held-out seed–cases from the four GP2PL-added families. Both rows use the same five evidence seeds; every valid plan passes VAL.

It remains separate from Table 1: that table compares compiler variants over all 16 domains, whereas this table compares provider execution with the final library on a paired scope. Published MOOSE coverage is near ceiling on its disjoint original scope (Chen et al. 2025); scope-separated MOOSE and per-instance planner references appear only in the Technical Supplement, Sec. 5.

8 Conclusion and Future Work

GP2PL demonstrates that generalized-planning evidence can be compiled into a maintained BDI plan library. Validated policy lifting selects an internally closed atomic core from evidence and typed PDDL schemas; DFA-guided compilation appends preservation-safe temporal controllers without relearning that core. Unsupported obligations fail closed rather than becoming unchecked plans.

Across 16 domains, five independently seeded cores achieved 98.68% mean held-out coverage (1.29-point sample SD), while recursive candidates improved paired atomic success by 10.42 percentage points. Preservation certification improved paired temporal success by 9.36 points; all 475 translations matched their gold finite-trace languages, and all 1,228 queries produced validated accepting traces. Certification improved both reusable coverage and temporally correct execution.

GP2PL does not claim arbitrary PDDL–LTL_f strategy synthesis: it rejects arithmetic outside the bounded-integer fragment, formulae outside the declared F, X, U, conjunction, and literal-negation fragment, and uncertified recursive or interference repair. Future work should broaden providers, candidate grammars, and temporal fragments under fail-closed certification, with pinned PDDL generators and held-out-disjoint instances for domain onboarding.

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